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Course Project Report

Thermochemical model fidelity in hypersonic nose drag, heating and shape

Course: Fundamentals of Hypersonic Flows

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Abstract: This work quantifies how the fidelity of the thermochemical model affects the aerothermodynamics and the minimum-drag shaping of a slender, axisymmetric hypersonic forebody. A body-conforming SU2/SU2-NEMO pipeline, validated against the Fay–Riddell correlation, compares a calorically perfect gas with a two-temperature, five-species non-equilibrium air model (Mutation++) on a Bézier ogive family at $M_\infty = 12$ and 45 km. The non-equilibrium model predicts a substantially lower stagnation heat flux—by about one third—driven by energy partition into internal modes and a frozen stagnation layer, while surface catalysis plays a minor role. Because this relaxes the heat-flux constraint, model fidelity shifts the constrained minimum-drag optimum towards sharper, lower-drag shapes, quantified here by a surrogate-based search.

1. Introduction

On a hypersonic nose, drag and heating are coupled and opposed. A sharp, slender nose minimises wave drag but drives the stagnation heat flux up, since $\dot{q}_s \propto R_n^{-1/2}$ [4, 11]; a blunt nose relieves the heating at the cost of pressure drag. Nose design is therefore a constrained optimisation: minimise drag subject to a survivable heat load.

1.1. State of the art

For bodies of revolution the drag-minimising profile at fixed length and base diameter is, by Newtonian theory, a power law $r(x) \propto x^n$ with $n \approx 0.7$ [3]; CFD places the optimum near $n \to 3/4$ over wide fineness and Mach ranges [8]. At fixed length and volume the Sears–Haack body and the von Kármán ogive are the closed-form wave-drag optima [1]. All are inviscid, perfect-gas results.

Two effects break those assumptions. (i) Viscous (skin-friction) drag, always a non-negligible share of the total and about a quarter to a third on a slender body, penalises wetted area and shifts the true optimum towards more slender shapes than inviscid theory predicts [2]. (ii) Behind a strong bow shock the air dissociates and its internal energy modes relax out of equilibrium; the two-temperature (T, T_{ve}) model [6] is the standard tractable approximation for this regime, and the wall heat flux then depends on surface catalysis through species recombination [10]. The Fay–Riddell [4] and

Sutton–Graves [11] correlations give the equilibrium stagnation heating but not these finite-rate effects. SU2, through its NEMO module [5] and the Mutation++ library [9], resolves both the perfect-gas and the two-temperature, five-species ($\mathrm{N}, \mathrm{O}, \mathrm{NO}, \mathrm{N}_2, \mathrm{O}_2$) flow over the same geometry, isolating the model effect.

1.2. Objective and scope

Classical shape-optimisation theory is inviscid and rests on the perfect-gas hypothesis, while the non-equilibrium heating literature rarely revisits the optimum. This work connects them, quantifying how thermochemical model fidelity affects (i) the aerothermodynamics—drag and stagnation heating—of a slender axisymmetric forebody and (ii) its minimum-drag shape under a heat-flux constraint.

The study is deliberately simplified: a two-dimensional axisymmetric, moderate-hypersonic point (specified in section 2.2), chosen for tractability. The body is thus not, a priori, a flight article; its value is methodological—the design guidelines are scale-invariant, and moving to a flight configuration is a computational, not a conceptual, step.

2. Physical and numerical modeling

2.1. Governing equations

Two model fidelities are compared on the same geometry. The *perfect-gas* level solves the compressible Navier–Stokes equations for a calorically perfect gas ($\gamma = 1.4$) with Sutherland viscosity. The *non-equilibrium* level uses the two-temperature (2-T) model [6]: the trans-rotational temperature T and the vibrational–electronic temperature T_{ve} are distinct, and air is a reacting mixture of five species ($\mathrm{N}, \mathrm{O}, \mathrm{NO}, \mathrm{N}_2, \mathrm{O}_2$). The conserved variables are the species densities ρ_s, the momentum $\rho\mathbf{u}$, the total energy ρE and the vibrational–electronic energy ρe_{ve}:

$$\begin{aligned} \partial_t \rho_s + \nabla\cdot(\rho_s \mathbf{u}) &= -\nabla\cdot(\rho_s \mathbf{V}_s) + \dot{\omega}_s, \\ \partial_t \rho\mathbf{u} + \nabla\cdot(\rho\mathbf{u}\otimes\mathbf{u} + p\mathbf{I}) &= \nabla\cdot\boldsymbol{\tau}, \\ \partial_t \rho E + \nabla\cdot[(\rho E + p)\mathbf{u}] &= \nabla\cdot(\boldsymbol{\tau}\cdot\mathbf{u} - \mathbf{q}), \\ \partial_t \rho e_{ve} + \nabla\cdot(\rho e_{ve}\mathbf{u}) &= -\nabla\cdot\mathbf{q}_{ve} + \dot{\omega}_{ve} + Q_{T\text{-}ve}. \end{aligned} \tag{1}$$

Here $\mathbf{V}_s$ is the species diffusion velocity, $\mathbf{q} = -k_{tr}\nabla T - k_{ve}\nabla T_{ve} + \sum_s \rho_s h_s \mathbf{V}_s$, $Q_{T\text{-}ve}$ the Landau–Teller vibrational–translational energy transfer, $\dot{\omega}_s$ the net chemical production rate of species s and $\dot{\omega}_{ve}$ its vibrational–electronic part. Reactions redistribute energy among the modes but conserve the total, so the ρE equation carries no chemical source. The thermodynamic, transport and chemical closures are supplied by Mutation++ [9]; the perfect-gas model is the inert, single-temperature limit.

Chemistry follows the Park air-5 mechanism: three dissociation reactions ($\mathrm{N}_2, \mathrm{O}_2, \mathrm{NO}$ + M) and two Zeldovich exchanges (N_2+O $\rightleftharpoons$ NO+N, NO + O $\rightleftharpoons$ O_2 + N), with T_{ve}-coupled dissociation rates [7].

2.2. Flight condition

The reference point is $M_\infty = 12$ at 45 km on the U.S. Standard Atmosphere (Table 1), with the free stream in thermal equilibrium ($T_{ve,\infty} = T_\infty$). At a body length $L = 0.1$ m the Reynolds number is $Re_L \approx 4.4 \times 10^4$, so the flow can be approximately considered laminar. The stagnation enthalpy ($h_0 \approx 7.9$ MJ/kg) is high enough to dissociate oxygen but, at this low density, too low to ionise appreciably—ionisation becomes significant only for $M \gtrsim 25$ [1]—so the five-species air model is adequate.

2.3. Wall thermochemistry

The wall is isothermal at $T_w = 300$ K. Catalysis is treated in three regimes: non-catalytic (no surface recombination), partially catalytic through a finite recombination efficiency γ (here $\gamma = 10^{-2}$), and super-catalytic (full recombination to the free-stream composition). In the γ-model the recombination mass flux of species s is $\dot{\omega}_s = \gamma_s Y_s \rho_w \sqrt{R_s T_w/(2\pi)}$ [5]. Recombination is exothermic and increases the wall heat flux [10]; the Fay–Riddell correlation captures this through the Lewis-number exponent (0.52

Table 1: Reference flight condition and derived quantities ($L = 0.1$ m).

Quantity	Value
Mach number M_∞	12
Altitude h	45 km
Temperature T_∞	265 K
Pressure p_∞	143 Pa
Density ρ_∞	1.9×10^{-3} kg/m^3
Velocity V_∞	3.9 km/s
Stagnation enthalpy h_0	7.9 MJ/kg
Reynolds number Re_L	4.4×10^4

for a non-catalytic/equilibrium wall, 0.63 for a fully catalytic one) [4].

3. Geometry and computational methodology

3.1. Shape parametrisation

The forebody is a smoothly blunted ogive described in the meridian plane by a single cubic Bézier curve (Figure 1),

$$\mathbf{B}(t) = \sum_{k=0}^{3} \binom{3}{k} (1-t)^{3-k} t^k \, \mathbf{P}_k, \qquad t \in [0,1], \tag{2}$$

with control points $\mathbf{P}_0 = (0,0)$, $\mathbf{P}_1 = (0, y_1)$, $\mathbf{P}_2 = (x_2, R_b)$ and $\mathbf{P}_3 = (L, R_b)$. Placing $\mathbf{P}_1$ on the axis gives a vertical leading tangent, which fixes the nose radius exactly: the apex curvature follows from (2) as

$$\kappa(0) = \frac{2x_2}{3\,y_1^2} = \frac{1}{R_n} \quad \Longleftrightarrow \quad y_1 = \sqrt{\tfrac{2}{3}\,R_n x_2}, \tag{3}$$

so $\kappa(0) = 1/R_n$ feeds the stagnation-heating correlation directly. The horizontal trailing tangent ($P_{2,y} = R_b$) makes the base parallel to the axis, and the curve is C^∞ with monotone radius and non-positive curvature by construction.

Three variables describe the family: the nose radius R_n, the fineness ratio $L/D = L/(2R_b)$ (so $L = (L/D)\,2R_b$), and a fullness parameter $c \in [0,1]$ that sets the second control abscissa

$$x_2 = (0.85 - 0.70\,c)\,L, \tag{4}$$

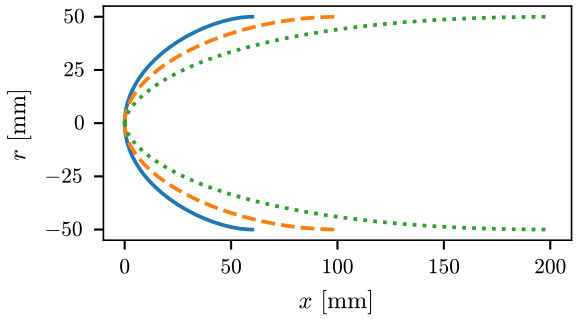


Figure 1: Representative members of the Bézier forebody family (solid: *blunt*, $R_n = 35$ mm, $L/D = 0.6$; dashed: *baseline*, $R_n = 20$ mm, $L/D = 1$; dotted: *slender*, $R_n = 5$ mm, $L/D = 2$; $R_b = 50$ mm). Varying R_n and L/D spans blunt-short to slender-long shapes.

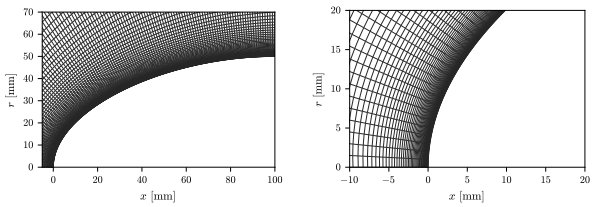
with $c \to 1$ giving a fuller ogive. The base radius R_b is held fixed and a shape is geometrically valid when $R_n x_2 \le 1.5\,R_b^2$; the baseline is $\{R_n = 0.02$ m, $c = 0.5$, $L/D = 1\}$ with $R_b = 0.05$ m. The design problem is

$$\min_{R_n,\,c,\,L/D} \; C_D \quad \text{s.t.} \quad \dot{q}_s \le \dot{q}_{\lim}, \tag{5}$$

with C_D referred to the frontal area πR_b^2 and $\dot{q}_s$ the stagnation heat flux.

3.2. Mesh and shock treatment

Since the wall heat flux $\dot{q}_w = k\,\partial T/\partial n$ is acutely sensitive to near-wall cell quality, the grid is a single-block, body-fitted structured mesh: a logically rectangular array of quadrilaterals mapped so that one family of grid lines follows the body surface and the other runs along the wall normals. It is generated by marching outward along those wall-normal lines, from the surface to an outer boundary that is a normal offset of the wall. The constraint placed on the cells is thus orthogonality at the wall together with a prescribed near-wall spacing (a cell Reynolds number of order unity)—precisely what makes $\dot{q}_w$ reliable. This is an H-type topology, not an O-grid wrapped around the nose: an O-grid would pinch to a singular cell at the stagnation point, whereas here the cells stay orthogonal at the wall and none degenerates (Figure 2). The offset grows from $1.5R_b$ at the tip to $4R_b$ at the base—a meshing choice, not a strict requirement, that keeps the bow shock inside the domain where it stands farther off downstream



(a) Full body (b) Leading-edge zoom

Figure 2: Body-conforming structured mesh (baseline): full body (a) and leading-edge zoom (b). Wall-normal marching keeps the cells orthogonal at the wall; the boundary layer is clustered to a cell Reynolds number of order unity.

while staying compact near the sharp tip. The base plane is closed with a characteristic (Riemann) far-field condition, suited to the mixed sub/supersonic outflow behind the bow shock (Figure 3).

Shock resolution uses a structured band re-clustering tailored to this body-conforming topology, rather than a general unstructured remesher, which would not preserve the structured wall-normal layout. Along each wall-normal grid line—a *column* running from the wall outward—let s be the distance travelled from the wall and d the column's total length to the far field, so the normalised coordinate $t = s/d$ runs from 0 at the wall to 1 at the outer boundary. The bow shock is detected on every column as the outermost point where the Mach number falls below a fixed fraction of the free stream, and the cells are re-clustered into the *same* normalised band $[t_1, t_2]$ on all columns; keying the band to t rather than to a fixed physical location keeps it aligned with the curved shock and precludes skew and folding. A graded node distribution keeps the physical first-cell height unchanged, so the boundary layer—and hence $\dot{q}_w$—is untouched while the shock is sharpened (Figure 4). The scheme is deliberately simple and not the most efficient: the refinement band is intentionally oversized, covering somewhat more than the shock itself. Adaptation is applied post-run—a converged coarse solution locates the shock, the grid is re-clustered, and the solver is restarted (warm) on the new grid. Each adapted cycle starts from the interpolated previous solution and therefore converges in comparatively few iterations: the cost

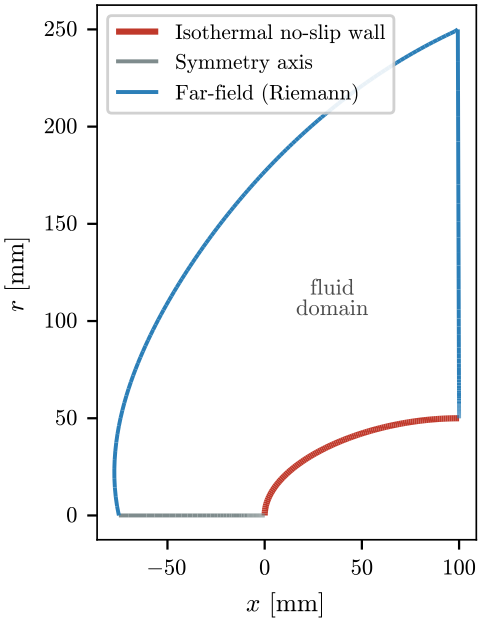


Figure 3: Computational domain and boundary conditions: isothermal no-slip wall, symmetry axis, and characteristic (Riemann) far-field closing both the outer boundary and the base.

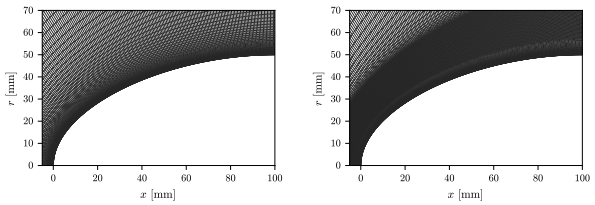
is essentially that of a second, short run.

3.3. Numerics

Both models are solved with SU2/SU2-NEMO [5]. The perfect-gas flow uses the AUSM convective scheme, weighted-least-squares gradients and implicit time integration with an adaptive CFL; first-order reconstruction (MUSCL disabled) is retained for robustness on the strong, grid-aligned bow shock. The non-equilibrium flow uses the carbuncle-robust AUSM$^+$-up2 scheme and explicit time integration (required by the Mutation++ coupling) at a reduced CFL. Convergence is monitored on the density and energy residuals and on the stationarity of the drag coefficient and integrated wall heat flux (Figure 5).

4. Baseline model comparison

The two model fidelities are compared on the baseline ogive ($R_n = 20$ mm, $c = 0.5$, $L/D = 1$) at the reference condition of section 2.2. The non-catalytic wall is used throughout, except in section 4.3.



(a) Baseline (b) After band adaptation

Figure 4: Structured band adaptation: (a) baseline grid and (b) after re-clustering, which sharpens the bow-shock region while leaving the wall-normal boundary-layer spacing unchanged.

Table 2: Perfect-gas stagnation heat flux $\dot{q}_s$ on the baseline ogive vs. the equilibrium correlations ($\dot{q}_{\mathrm{FR}} = 3.26$, $\dot{q}_{\mathrm{SG}} = 3.21$ MW/m^2).

Grid	$\dot{q}_s$ [MW/m^2]	error vs. FR
Coarse	2.82	−13.7%
Fine	3.08	−5.5%

4.1. Perfect-gas validation

The perfect-gas computation anchors the methodology against the Fay–Riddell [4] and Sutton–Graves [11] stagnation-heating correlations. Fay–Riddell is a semi-empirical estimate of the stagnation-point heat flux of a blunt body with an equilibrium boundary layer; it scales as $\dot{q}_s \propto \sqrt{\mathrm{d}u_e/\mathrm{d}x}\,(h_0 - h_w)$, in which the stagnation velocity gradient is fixed by the nose radius. It therefore depends on the local bluntness and the edge state, not on the afterbody: the present ogive has an exact apex radius $R_n = \kappa(0)^{-1}$ (section 3.1), so the correlation applies at its stagnation point and the baseline shape needs no modification for the comparison.

The computed stagnation heat flux approaches the correlations as the grid is refined (Table 2). Here "fine" denotes a smaller first-cell height at the wall: it is the near-wall spacing, not the overall cell count, that controls $\dot{q}_w = k\,\partial T/\partial n$, because SU2 imposes the wall temperature *weakly*—the energy flux is read from the discrete normal gradient, so the near-wall T approaches T_w only as the first cell shrinks. The grid-convergent agreement (within 5.5% on the fine grid) validates the near-wall resolution and the heat-flux evaluation on which the rest of the study depends.

4.2. Non-equilibrium effect

Replacing the perfect gas with the two-temperature five-species model on the *same* grid lowers the stagnation heat flux by about one third (2.82 → 1.91 MW/m^2, −32%; Figure 9), while the drag coefficient is essentially identical—$C_D = 0.513$ (perfect gas) versus 0.510 (air-5), a difference below 1%: model fidelity reshapes the heating, not the drag. Two mechanisms combine. First, the post-shock enthalpy is partitioned across internal modes: behind the bow shock the trans-rotational temperature runs well ahead of the lagging vibrational–electronic temperature (Figures 6–7), a strong thermal non-equilibrium that withholds energy from the translational pool driving wall conduction. Second, at this low density the chemistry is largely frozen (Figure 8): immediately behind the shock the gas is hot but still undissociated, dissociation then builds to a peak of only $\approx 6\%$ atomic oxygen by mass toward the stagnation point, and these atoms fail to recombine across the cooled boundary layer—Y_{O} is still $\approx 5\%$ at the cold wall, where equilibrium would recombine it entirely. Little dissociation enthalpy is therefore released at the non-catalytic wall. The same frozen state explains why the absolute air-5 heat flux falls well short of Fay–Riddell, which presumes an equilibrium boundary layer that the present shock layer does not attain.

4.3. Surface catalysis

Wall catalysis is a second-order effect at this condition. Across the three regimes the stagnation heat flux follows the expected ordering—non-catalytic < γ-model < super-catalytic—but spans less than 10% (Table 3): with so few atoms reaching the cold wall, surface recombination releases little additional energy. Catalytic modelling, often decisive at lower altitude where the boundary layer is closer to equilibrium, is thus not a leading-order uncertainty here.

This spread admits a simple numerical bracket. From the stagnation-line composition (Figure 8) the chemical enthalpy stored in dissociation at the boundary-layer edge is $h_D = \sum_i Y_i\,\Delta h^\circ_{f,i} \approx 1.0$ MJ/kg (dominated by atomic oxygen), i.e. a fraction $h_D/h_0 \approx 0.13$ of the stagnation enthalpy $h_0 \approx 7.9$ MJ/kg. A super-catalytic wall can recover at most this fraction, so the catalytic enhancement is bounded above by $\approx 13\%$;

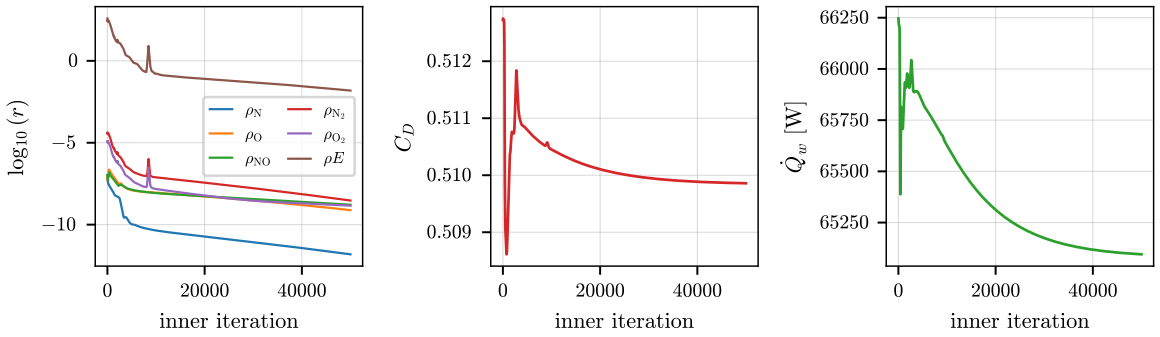
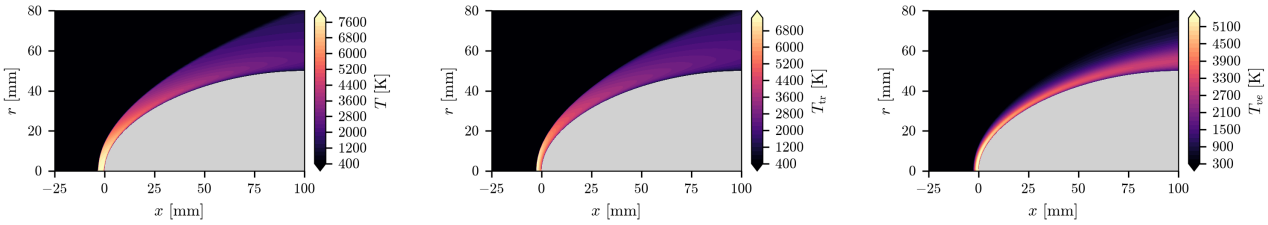


Figure 5: Convergence history of an air-5 run: per-species density and energy residuals (left), drag coefficient (centre) and integrated wall heat flux $\dot{Q}_w$ (right). The adapted cycle is warm-started, so the residual restarts at a low level and settles in relatively few iterations.



(a) Perfect gas, T (b) Air-5, T_{tr} (c) Air-5, T_{ve}

Figure 6: Baseline ogive, perfect gas vs. two-temperature air-5 (non-catalytic): (a) perfect-gas temperature; (b,c) air-5 trans-rotational and vibrational–electronic temperatures, whose large separation shows the post-shock partition of energy into internal modes.

Table 3: Stagnation heat flux on the baseline ogive (matched coarse grid) for the three wall catalysis regimes; $\dot{q}_{\mathrm{nc}}$ is the non-catalytic value.

Wall regime	$\dot{q}_s$ [MW/m^2]	$\dot{q}_s/\dot{q}_{\mathrm{nc}}$
Non-catalytic	1.91	1.00
γ-model (10^{-2})	1.95	1.02
Super-catalytic	2.09	1.09

the Fay–Riddell Lewis-number weighting of that fraction, $(\mathrm{Le}^{0.63}-1)\,h_D/h_0$ with $\mathrm{Le}\approx 1.4$, sharpens the estimate to $\approx 3\%$. The CFD value, +9% (Table 3), sits inside this 3–13% band—the same single-digit order, and far below the near-doubling reached in high-enthalpy equilibrium flows. This is only a consistency check: Fay–Riddell presumes an equilibrium boundary layer, whereas the present stagnation layer is frozen.
In summary, model fidelity changes the stagnation heat flux by a third while leaving the drag almost untouched. In the constrained design problem (5) it is therefore the heat-flux constraint, not the objective, that moves—which is what makes the minimum-drag optimum model-dependent (section 5).

5. Shape optimization and model dependence

The design problem (5) is solved by surrogate-based constrained optimisation over the three shape variables $\{R_n,\, c,\, L/D\}$. The heat limit is set, arbitrarily, to the Fay–Riddell reference value $\dot{q}_{\mathrm{lim}} = 3.26$ MW/m^2—a convenient anchor, not a physical survivability threshold, so the optimum below should be read relative to it. A space-filling Latin-hypercube design of 15 points samples the valid envelope ($R_n x_2 \le 1.5R_b^2$, by rejection) across the three design variables (Table 4). Each sample is evaluated by the pipeline of section 3 on a coarse mesh, returning the drag coefficient C_D and the stagnation heat flux $\dot{q}_s$. The campaign is run on the perfect-gas model and without shock adaptation, deliberately, to keep it affordable: the

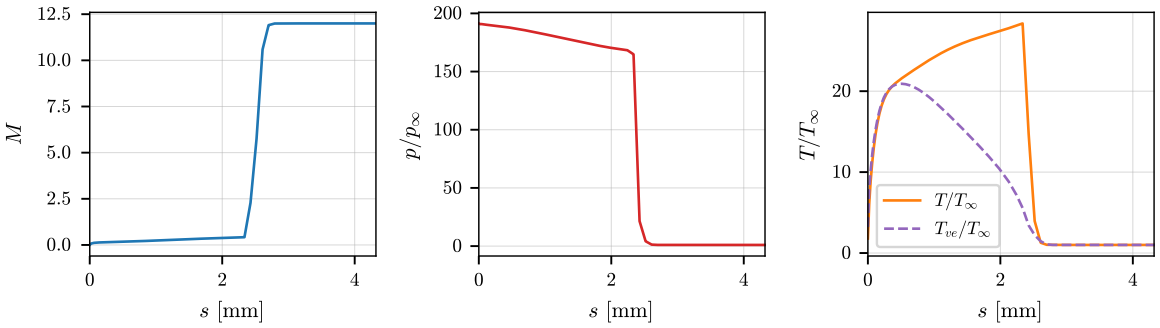


Figure 7: Stagnation-line profiles (air-5, non-catalytic) from the wall to the shock: Mach number, pressure and the two temperatures. The large separation $T_{\mathrm{tr}} \gg T_{ve}$ is the signature of the frozen, thermally non-equilibrium stagnation layer.

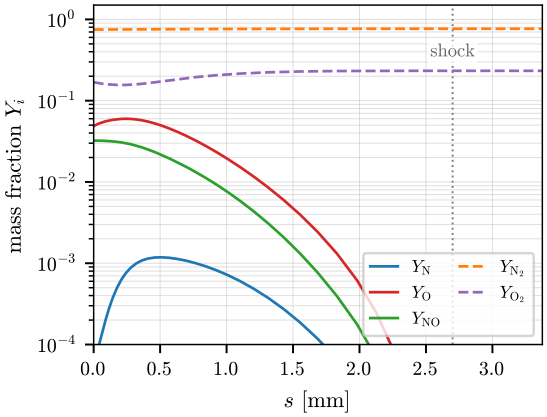


Figure 8: Stagnation-line species mass fractions (air-5, non-catalytic) from the wall ($s = 0$) to the bow shock. Atomic oxygen produced behind the shock peaks at $\approx 6\%$ near the stagnation point and remains near 5% at the cold wall—where equilibrium would recombine it entirely—the signature of a chemically frozen shock layer.

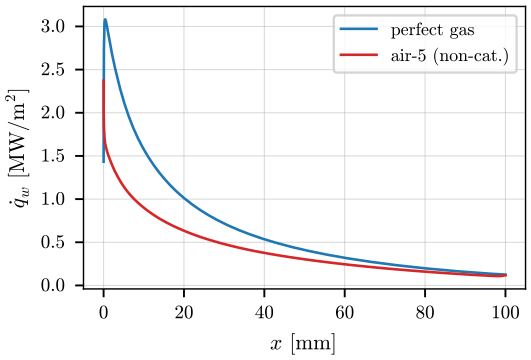


Figure 9: Wall heat flux $\dot{q}_w(x)$ along the baseline ogive: the two-temperature air-5 model lies below the perfect gas over the whole surface, by about one third at the stagnation point.

perfect-gas C_D is already grid-converged on the coarse mesh (section 3.2), so adaptation buys nothing here. This is a matter of economy, not capability: the pipeline, the band adaptation and the air-5 model all apply unchanged to a higher-fidelity campaign, only at substantially higher cost. Two independent Gaussian-process (Kriging) surrogates are then fitted, one for C_D and one for $\dot{q}_s$, and 20 infill iterations maximise the constrained expected improvement—expected improvement weighted by the probability that $\dot{q}_s \leq \dot{q}_{\mathrm{lim}}$—which steers the search to the feasible frontier without tuning a penalty coefficient.

The 35 evaluations populate the design space and the infill points concentrate along the heat-flux frontier, where the feasible minimum-drag shape lies (Figure 10). Under the perfect-gas constraint the optimiser returns a slender body (Table 5): the drag falls about 45% below the baseline $C_D \simeq 0.51$ while the stagnation heat flux sits exactly on the constraint. The optimum sits at the most slender shape the search allows (Figure 11): L/D rests at the upper bound of the box while the heat flux saturates the limit, so both the slenderness cap and the thermal constraint are active. Under perfect-gas heating it is thus the wave-drag penalty, not the heating, that drives the shape—and a wider L/D range would push the optimum further still.

The constraint that shapes the optimum is exactly the quantity that model fidelity moves. Section 4 showed air-5 predicting a stagnation heat flux about a third lower than the perfect gas at the same shape, with the drag essentially

Table 4: Design space and sampling budget of the surrogate campaign (perfect gas).

Quantity	Value
Nose radius R_n	5–35 mm
Fullness c	0.2–1.0
Fineness L/D	0.6–3.0
Base radius R_b (fixed)	50 mm
Heat limit $\dot{q}_{\mathrm{lim}}$	3.26 MW/m^2
Samples (LHS + infill)	15 + 20

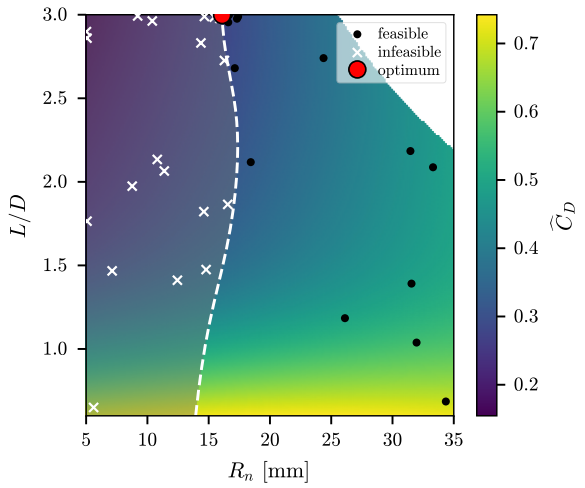


Figure 10: Kriging surrogate of the drag coefficient $\widehat{C}_D$ over the $(R_n, L/D)$ plane at $c = c_{\mathrm{opt}}$. Dashed: the heat-flux constraint $\dot{q}_s = \dot{q}_{\mathrm{lim}}$ (grey/white: heat- or shape-infeasible). The marker is the constrained minimum-drag optimum, on the frontier at the largest L/D; the 35 samples are overlaid (dots feasible, crosses infeasible).

model-independent. A spot-check at the perfect-gas optimum confirms this (Table 5): the air-5 heat flux there, $\dot{q}_s \approx 2.51$ MW/m^2, sits below the limit, so the non-equilibrium constraint is slack where the perfect-gas one is active. A perfect-gas design thus leaves margin that the higher-fidelity model would spend on a sharper, lower-drag nose, and the viscous skin-friction penalty pushes the optimum the same way [2]. Re-running the model-agnostic loop under air-5 would quantify the full shift; here it is bounded by the spot-check.

Table 5: Constrained minimum-drag optimum: shared design variables and the resulting drag and heating per model (air-5 is a spot-check at the perfect-gas optimum, where lower non-equilibrium heating relaxes the active constraint).

R_n [mm]	c	L/D
16.1	0.52	3.0

	Perfect gas	Air-5 (spot-check)
C_D	0.277	0.281
$\dot{q}_s$ [MW/m^2]	3.26	2.51

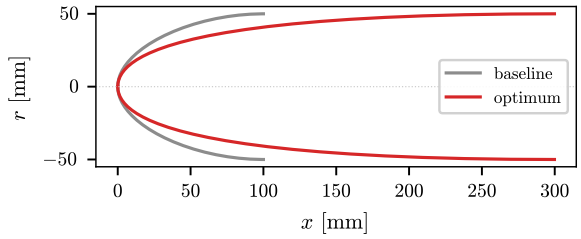


Figure 11: Meridian silhouette of the optimum (Table 5) vs. the baseline ($R_n = 20$ mm, $c = 0.5$, $L/D = 1$): sharper and three times longer, trading wetted area for lower wave drag within the heat limit.

6. Conclusions

This work quantified the effect of thermochemical model fidelity on the aerothermodynamics and minimum-drag shaping of a slender axisymmetric hypersonic forebody, using a structured, body-conforming SU2/SU2-NEMO pipeline validated against the Fay–Riddell stagnation-heating correlation. Three points stand out. (i) Relative to a calorically perfect gas, the two-temperature five-species model predicts a substantially lower stagnation heat flux—by about one third on the matched grid—because the shock-layer energy is partitioned into vibrational and chemical modes and, at this low-density condition, the chemistry is largely frozen. (ii) Surface catalysis has only a minor effect here: the ordering non-catalytic $<$ γ-model $<$ super-catalytic holds but spans a few per cent, again because the stagnation layer is frozen. (iii) Because the non-equilibrium model relaxes the heat-flux constraint, the constrained minimum-drag optimum is model-dependent:

under the perfect-gas constraint the surrogate campaign returns a slender body ($L/D \approx 3.0$, $C_D \approx 0.277$, against 0.51 at the baseline), and the lower non-equilibrium heating leaves margin for a still sharper, lower-drag nose. The analysis is intentionally simplified, but the methodology and the trade-off it exposes—drag against a heat-flux constraint that the thermochemical model controls—carry over unchanged to higher-fidelity models and flight conditions.

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